

AyurCentral HRMS — Full documentation (reference PDF)

Product: AyurCentral Enterprise HRMS
Stack: Google Apps Script web app + one Google Spreadsheet + Google Drive
Branch reference: `cursor/experiment-500f` (two-stage leave, bulk approvals)

1. What this system is

AyurCentral HRMS is a single-tenant HRMS for employees, leave, payroll, recruitment (ATS), notifications, and an **Ask HR** knowledge assistant. All authoritative data lives in **Google Sheets** tabs; files (payslips, employee documents) live in **Google Drive**. Users sign in with **Google email + OTP**; authorization is enforced on every server API via `PermissionService`.

2. Modules at a glance

Module	Purpose	Main UI routes
Foundation	Auth, permissions, DB, config, dashboard shell	Sign-in, Dashboard
Employee	Directory, profiles, bulk upload, documents	<code>employees</code> , <code>my-profile</code> , <code>my-team</code>
Leave	Apply, multi-stage approvals, types, balances	<code>my-leave</code> , <code>leave-approvals</code> , <code>leave-admin</code>
Payroll	Runs, inputs, calculate, lock, payslips	<code>payroll</code> , <code>my-payslips</code>
Compensation	Salary structures & components	<code>compensation</code> (often under Payroll nav)
Notifications	In-app bell, preferences, email log	<code>notifications</code>
Ask HR	RAG-style HR Q&A (Knowledge Hub)	Floating panel on all pages
ATS (Recruitment)	Jobs, candidates, applications, interviews	<code>ats</code> , <code>ats-jobs</code> , <code>ats-candidates</code>

3. Roles and navigation (summary)

Role	Typical access
OWNER / ADMIN	Setup, users (placeholders), all HR modules, final leave stages, payroll
HR	Employees, leave admin, payroll, recruitment, notifications
MANAGER	My team, stage-1 leave approval for direct reports, own leave/payslips, ATS
EMPLOYEE	My profile, my leave, my payslips (if <code>Users.access_*</code> flags allow)

Detailed matrix: see `03_USER_ROLES_PERMISSIONS.md`.

4. Leave workflow (current policy)

4.1 Who approves whom

Applicant (<code>Users.role</code>)	Approval chain
Employee (with manager)	Manager → then HR or Admin (either one completes stage 2)
Employee (no manager)	HR or Admin (one step)
Manager	HR → then Admin (both required, in order)
HR	Admin only

Statuses on **LeaveRequests**: `PENDING_MANAGER`, `PENDING_HR`, `PENDING_ADMIN`, then `APPROVED` / `REJECTED` / `CANCELLED`. Legacy `SUBMITTED` is treated as `PENDING_MANAGER`.

4.2 Balances

- **Leave balances are not in the employee CSV.** On create/bulk upload, rows are seeded on **LeaveBalances** from **LeaveTypes** `annual_entitlement_days`.
- **New leave year:** Leave admin → **Start leave year (grant balances)**.

4.3 Bulk approval (UI)

On **Leave Approvals** and **Leave admin** → **Requests: Select all**, row checkboxes, **Approve selected** / **Reject selected** (one comment for the batch). Server runs the same rules per request.

5. Data storage (quick index)

Sheet tab	Primary key	Purpose
Employees	employee_id	Person master
Users	google_email	Login, role, access flags
LeaveTypes / LeaveBalances / LeaveRequests	various	Leave policy & workflow
SalaryStructures / SalaryComponents	various	Compensation
PayrollRuns / PayrollInputs / PayrollRecords	various	Monthly payroll
Documents	document_id	Drive file metadata
Notifications	notification_id	Email send log
NotificationInbox / NotificationPreferences	various	In-app bell
AuditLog	audit_id	Audit trail
Settings	setting_key	Config + ID sequences
JobRequisitions, Candidates, Applications, Interviews, CandidateActivity	ATS keys	Recruitment

Full field-level map: `docs/HRMS_COMPLETE_DATA_AND_FEATURES.md` (in workspace) or spreadsheet spec `02_DATABASE_SCHEMA.md`.

Not in Sheets: OTP and session tokens in **CacheService**.

6. Employee lifecycle (short)

- **Create:** Employees row + optional Users row + auto **LeaveBalances** + Drive folder + audit.
- **Deactivate:** `Employees.status` = INACTIVE, `Users.status` = DISABLED — history retained.
- **No app “delete person”** — do not delete sheet rows manually without fixing dependents.

7. Deploy checklist

1. `git pull origin cursor/experiment-500f`
2. `cd apps-script` → `clasp push` (folder with `.clasp.json`, not a nested clone)
3. Apps Script → **Deploy** → **Manage deployments** → **New version**
4. Hard refresh the web app URL

Appendix A — Repository documentation files

Each file below lives in the **ayurcentral-hrms** Git repo. Use it for product, data, or implementation detail.

File	What it is
00_MASTER_SPEC.md	Top-level product scope and document index for the spec set
01_PRODUCT_REQUIREMENTS.md	MUST/SHOULD requirements (notifications, payroll states, etc.)
02_DATABASE_SCHEMA.md	Canonical column list for every sheet tab
03_USER_ROLES_PERMISSIONS.md	Roles, nav matrix, permission intent
04_EMPLOYEE_MODULE.md	Employee CRUD, profile, documents behavior
05_LEAVE_MODULE.md	Leave states, validation, balance rules (update with multi-stage in code)
06_PAYROLL_COMPENSATION_MODULE.md	Payroll run workflow and compensation structures
07_DASHBOARD_NOTIFICATIONS.md	Dashboard widgets and MailApp notification events
08_UI_DESIGN_SYSTEM.md	UI patterns, components, UX conventions
09_APPS_SCRIPT_ARCHITECTURE.md	Layers, services, concurrency, no Advanced APIs
10_SECURITY_AUDIT.md	Security assumptions and review notes
11_TEST_PLAN.md	Test strategy and P0 cases
12_BUILD_PLAN.md	Build/phasing notes
apps-script/SETUP.md	clasp, script properties, first-time setup
docs/ATS_INTEGRATION_NOTES.md	How ATS connects to HRMS sheets and Drive
docs/BULK_UPLOAD_TESTING_GUIDE.md	Testing employee/payroll bulk uploads
docs/HRMS_GAP_ANALYSIS.md	Known gaps vs requirements
docs/NOTIFICATIONS_INTEGRATION_NOTES.md	Wiring leave/payroll events to NotificationEngine
docs/PERF_BASELINE.md	Performance baseline measurements
docs/SHEETS_PERFORMANCE_AUDIT.md	Spreadsheet read/write patterns
docs/HRMS_Bulk_Employee_Upload_Template.csv	CSV template for employee bulk upload
docs/HRMS_FULL_DOCUMENTATION.md	This document (source for PDF)

Appendix B — Apps Script project files (module & feature)

In the Apps Script editor, files appear with a **folder prefix** (e.g. `ats/ApiAts.gs`). Below, **Module** = product area; **Feature** = what the file does.

B.1 Project root

File	Module	Feature
<code>appsscript.json</code>	Project	Manifest (time zone, web app, OAuth scopes)

B.2 foundation/ — Core platform

File	Feature
<code>Main.gs</code>	Web app <code>doGet</code> , HTML <code>include()</code> , spreadsheet menu (setup, ATS/notification schema)
<code>Constants.gs</code>	Sheet names, roles, actions, employee/user status enums
<code>ApiFoundation.gs</code>	Auth OTP APIs, home dashboard bundle, lazy module UI loader, setup runners
<code>AuthService.gs</code>	Login, OTP issue/verify, session resolution
<code>AuthServiceService.gs</code>	Session token storage and validation helpers
<code>PermissionService.gs</code>	Role-based actions and sidebar navigation
<code>UserAccessService.gs</code>	Per-user flags: <code>access_leave</code> , <code>access_payslips</code> , <code>access_documents</code>
<code>DbService.gs</code>	Spreadsheet CRUD, sequences, batch insert
<code>SchemaService.gs</code>	Bootstrap sheet headers and setup
<code>ConfigService.gs</code>	Settings key/value with cache
<code>AuditService.gs</code>	Append audit log rows
<code>Errors.gs</code>	Typed errors (validation, auth, conflict, not found)
<code>DriveService.gs</code>	HRMS Drive root and employee document folders
<code>HomeDashboardService.gs</code>	Role-scoped dashboard counts (leave queue, payroll, ATS, notifications)
<code>LockUtil.gs</code>	Script lock helpers
<code>PerfUtil.gs</code>	<code>HrmsPerf</code> timing for slow-path investigation

B.3 employee/ — People & directory

File	Feature
EmployeeService.gs	Create/update/status, validation, leave balance seed on hire, Drive folder
EmployeeRepository.gs	Sheet access for Employees, Users, leave types/balances lookups
EmployeeBulkService.gs	Validate/commit bulk employee CSV upload
ApiEmployee.gs	Client APIs: list, get, create, update, status
ApiEmployeeBulk.gs	Bulk upload validate/commit APIs
ApiEmployeeAccess.gs	Document access APIs for employee files
EmployeeClient.html	Employees UI: directory, create, profile client logic
EmployeePages.html	Employee page templates/partial

B.4 leave/ — Time off

File	Feature
LeaveEngine.gs	Pure rules: day count, overlap, multi-stage approval auth, status labels
LeaveService.gs	Types, balances, submit/approve/reject/cancel/revoke, bulk approve/reject, calendar, year start
LeaveLopService.gs	LOP days from approved leave for open payroll runs
ApiLeave.gs	All <code>apiLeave*</code> endpoints for client
LeaveUi.html	Leave screens: apply, my leave, approvals (incl. bulk select), admin, calendar
LeaveClient.html	Leave module registration and route hooks

B.5 payroll/ — Payroll & compensation

File	Feature
PayrollEngine.gs	Pure pay calculation from structure + inputs
PayrollService.gs	Runs lifecycle: draft, calculate, review, approve, lock
PayrollBulkService.gs	Bulk attendance / payroll input upload
PayrollLeaveBridge.gs	Refresh LOP from leave on unlocked runs
CompensationService.gs	Salary structure CRUD and components
CompensationBulkService.gs	Bulk compensation upload

File	Feature
PayslipService.gs	Payslip generation and document records
ApiPayroll.gs	Client payroll and compensation APIs
PayrollClient.html	Payroll UI: runs, inputs, bulk sections, payslips

B.6 notifications/ — Notification center

File	Feature
NotificationEngine.gs	Event catalog, payloads, routing (leave, payroll, ATS)
NotificationService.gs	Create inbox rows, email via MailApp, preferences
NotificationSchema.gs	Ensure NotificationInbox / Preferences sheets
NotificationAdapters.gs	<code>NotificationLeaveAdapter</code> etc. — called after business commits
NotificationTriggers.gs	Time-driven or batch trigger helpers
ApiNotifications.gs	Bell, inbox, preferences, announcements APIs
NotificationClient.html	Notifications page UI
NotificationBell.html	Header bell dropdown

B.7 knowledge/ — Ask HR

File	Feature
AskHrService.gs	Orchestrates question → Knowledge Hub / grounding
KnowledgeHubClient.gs	HTTP client to external Knowledge Hub web app
AskHrEmployeeLookup.gs	Optional employee context for answers
AskHrDiagnostic.gs	Admin/diagnostic helpers for Ask HR setup
ApiAskHr.gs	<code>apiAskHr*</code> client endpoints

B.8 ats/ — Recruitment (ATS)

File	Feature
AtsConstants.gs	ATS sheet names and enums
AtsSchemaService.gs	ATS tab headers and ensure schema
AtsRepository.gs	CRUD for jobs, candidates, applications, interviews

File	Feature
AtsEngine.gs	Business rules and permissions helpers
AtsPermissionService.gs	ATS-specific role checks
AtsService.gs	Jobs, candidates, pipeline, interviews orchestration
AtsBulkService.gs	Bulk import for candidates/applications
AtsDriveService.gs	Resume and attachment folders on Drive
ApiAts.gs	Authenticated ATS APIs for HRMS UI
ApiAtsBulk.gs	Bulk ATS APIs
ApiAtsPublic.gs	Public job board / apply endpoints (unauthenticated)
AtsWeb.gs	Public <code>doGet</code> routing for careers/apply URLs
AtsApp.html	ATS shell / layout in Apps Script
AtsClient.html	ATS module UI (jobs, candidates, kanban-style flows)
AtsPublicApply.html	Public application form HTML

B.9 ui/ — HRMS shell (all modules)

File	Feature
Index.html	Main SPA shell: auth gate, sidebar, route container
Scripts.html	Router, <code>callServer</code> , nav groups, dashboard, module loader
Styles.html	Global CSS, design tokens, tables, bulk toolbars

B.10 tests/ — Apps Script unit tests (run in editor)

File	Feature
TestFoundation.gs	Schema, auth, permissions smoke tests
TestEmployee.gs	Employee service tests
TestLeave.gs	Leave workflow tests
TestPayroll.gs	Payroll engine/service tests
TestAts.gs	ATS tests
TestNotifications.gs	Notification tests
TestAskHr.gs	Ask HR tests

File	Feature
TestAuthOtp.gs / TestAuthDemo.gs	Auth flows
TestStartupPerf.gs / TestPerfInvestigation.gs	Performance
TestWebAppExecution.gs	Web app smoke

B.11 Repo tests/ (Node, on your PC)

Node tests under `tests/*.test.js` validate HTML/ `.gs` contracts (leave, payroll, nav, notifications) without deploying. Run: `node tests/leave-engine.test.js`.

Appendix C — How files map to the Apps Script editor

- **clasp** pushes `apps-script/src/**` to the cloud project; folder names become the prefix you see (e.g. `ats/ApiAts.gs`).
- **HTML** files in `ui/`, `leave/`, `employee/`, etc. are included via `include()` or lazy-loaded through `apiGetModuleUi`.
- **Entry point** for employees using HRMS: `foundation/Main.gs` → `ui/Index.html`.
- **Entry point** for public careers pages: `ats/AtsWeb.gs` + `ats/AtsPublicApply.html` when configured.

Generated for AyurCentral HRMS operators and implementers. For the widest data dictionary, also keep `HRMS_COMPLETE_DATA_AND_FEATURES.md` in your docs folder.